

Q1 Period of $\cos 3x$ is $2\pi/3$ Period of $\cos x = 2\pi$

Q2 If $x=c$ is a point of discontinuity then the Fourier Series of $f(x)$ at $x=c$ gives $f(c) = \frac{1}{2} [f(c-0) + f(c+0)] = \frac{1}{2} [f_c^-(c) + f_c^+(c)]$

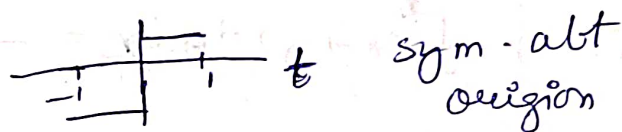
Q3 Fourier expansion of an odd f^n has only sine terms.

Q4 If $f(x) = x^4$ in $(-1, 1)$ then the Fourier coefficient $b_n =$ 0.

Q5 constant f^n is periodic True/False.

Q6 The period of a constant f^n is not defined.

Q7. If $f(t) = \begin{cases} -1 & -1 < t < 0 \\ 1 & 0 < t < 1 \end{cases}$ then $f(t)$ is an odd f^n



Q8 The smallest period / fundamental period of the f^n $\sin\left(\frac{2n\pi x}{k}\right)$ is $2\pi / (2n\pi/k) = k/n$

Q9 In the Fourier series expansion of $f(x) = |\sin x|$ in $(-\pi, \pi)$ the value of $b_n =$ 0.
 $\sin x$ — odd. but $|f(x)|$ is always even.
 $\ast$ modulus of a f^n is always even.

Q10 In the Fourier series for $f(x) = x$ in $(-\pi \leq x \leq \pi)$, the cosine terms are absent

Q11 If $f(x) = x^2$ in $(-2 < x < 2)$ then a_n is $\frac{2}{\pi} \int_0^2 x^2 \cos\left(\frac{n\pi x}{2}\right) dx$.

$l=2$ As x^2 is an even f^n .

Q12 If $f(x) = x \sin x$ in $(-\pi, \pi)$ then the values of b_n is zero.

x — odd
 $\sin x$ — odd
 $\text{odd} \times \text{odd} = \text{even } f^n$

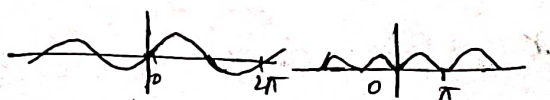
Q13 The half range sine series for 1 in $(0, \pi)$ is

$$\frac{2}{\pi} \int_0^\pi \sin nx \, dx = \frac{-2}{\pi} \left[\frac{\cos nx}{n} \right]_0^\pi = \frac{2}{n\pi} (1 - (-1)^n) = \frac{2}{n\pi} (1 - (-1)^n)$$

$$\frac{2}{\pi} \text{ Fourier series } = \frac{2}{\pi} (1-1) \sin x + \frac{2}{\pi} (1-(-1)) \sin 2x + \frac{2}{\pi} (1-1) \sin 3x + \dots$$

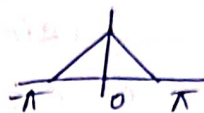
$$= \frac{4}{\pi} \left(\sin x + \frac{\sin 3x}{3} + \frac{\sin 5x}{5} + \dots \right)$$

Q14 Period of $|\sin t|$ is π



Q15 $f(x) = \begin{cases} -x & -\pi < x < 0 \\ x & 0 < x < \pi \end{cases}$ then $f(0) = \underline{-\pi/2}$
 As 0 is the pt. of discont. so.
 $f(0) = \frac{1}{2} [f(0^+) + f(0^-)]$
 $= \frac{1}{2} [0 + (-\pi)] = -\pi/2$

Q16 The fn $f(x) = \begin{cases} 1-x & \text{in } -\pi < x < 0 \\ 1+x & \text{in } 0 < x < \pi \end{cases}$ is an odd fn
 True / False



Q1 $\int_0^2 \int_0^x (x+y) dy dx = \underline{4}$ $\int_0^2 xy + y^2/2 \Big|_0^x = \int_0^2 \frac{x^2 + x^2}{2} dx = \frac{x^3 + x^3}{2} \Big|_0^2 = \frac{1}{2}(8) = 4$

Q2 $\int_0^1 \int_0^{1-x} dy dx = \underline{1/2}$ $\int_0^1 (1-x) dx = x - \frac{x^2}{2} \Big|_0^1 = 1 - \frac{1}{2} = \frac{1}{2}$
 It gives you the area of a triangle.
 $y = 1-x$

Q3 $\int_0^2 \int_1^3 \int_1^2 xy^2z dz dy dx = \int_0^2 x dx \int_1^3 y^2 dy \int_1^2 z dz$
 $= \frac{x^2}{2} \Big|_0^2 \cdot \frac{y^3}{3} \Big|_1^3 \cdot \frac{z^2}{2} \Big|_1^2 = (2) \left(\frac{27}{3} - \frac{1}{3} \right) \left(\frac{2}{2} - \frac{1}{2} \right)$
 $= (2) \left(\frac{26}{3} \right) \left(\frac{1}{2} \right) = \underline{26}$

Q4 $\iint x^2 y^3 dx dy$ over the rectangle $0 \leq x \leq 1$
 & $0 \leq y \leq 3$ is $\underline{27/4}$

Q5 $\int_0^\pi \int_0^{a \sin \theta} r dr d\theta = \underline{\pi a^2/12}$

Q6 $\int_0^1 \int_x^{\sqrt{x}} (x^2 + y^2) dx dy = \underline{-1/3}$

Q7 Area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 = ab\pi$

Q8 $\iint (x^2 + y^2) dx dy$ in the positive quadrant for which $x+y \leq 1$ is $\underline{1/6}$

Q9 $\iint dx dy$ over the area bounded by $x=0, y=0, x^2+y^2=1$ & $5y=3$ is $\underline{\frac{6}{25} + \frac{1}{2} \sin \frac{3}{5}}$

Q10 $\iint xy (x+y) dx dy$ over the area between $y=x^2$ & $y=x$ is $\underline{3/256}$